

this was irrelevant and that the relevant measure of risk was how the asset co-varied with the market portfolio—the beta of the asset. It turns out that asset volatility itself matters, as we shall see in chapter 7, but for the purpose of describing the CAPM and its incredible implications, we can ignore this for the time being.

The CAPM was formulated in the 1960s by Jack Treynor (1961), William Sharpe (1964), John Lintner (1965), and Jan Mossin (1966), building on the principle of diversification and mean-variance utility introduced by Harry Markowitz in 1952. For their work on CAPM and portfolio choice, Sharpe and Markowitz received the 1990 Nobel Prize in economics. (Merton Miller was awarded the Nobel Prize the same year for contributions to corporate finance.) Lintner and Mossin, unfortunately, had both died by then. Treynor, whose original manuscript was never published, has never received the recognition that he deserved.

I state upfront that the CAPM is well known to be a spectacular failure. It predicts that asset risk premiums depend only on the asset's beta and there is only one factor that matters, the market portfolio. Both of these predictions have been demolished in thousands of empirical studies. But, the failure has been glorious, opening new vistas of understanding for asset owners who must hunt for risk premiums and manage risk.

The basic intuition of the CAPM still holds true: that the factors underlying the assets determine asset risk premiums and that these risk premiums are compensation for investors' losses during bad times. Risk is a property not of an asset in isolation but how the assets move in relation to each other. Even though the CAPM is firmly rejected by data, it remains the workhorse model of finance: 75% of finance professors advocate using it, and 75% of CFOs employ it in actual capital budgeting decisions despite the fact that the CAPM does not hold.¹ It works approximately, and well enough for most applications, but it fails miserably in certain situations (as the next chapter will detail). Part of the tenacious hold of the CAPM is the way that it conveys intuition of how risk is rewarded.

What does the CAPM get right?

3.1. CAPM LESSON 1: DON'T HOLD AN INDIVIDUAL ASSET, HOLD THE FACTOR

The CAPM states that one factor exists and that factor is the market portfolio, where each stock is held in proportion to its market capitalization. This corresponds to a market index fund.² The factor can be optimally constructed by holding many assets so that nonfactor, or idiosyncratic risk, is diversified away. Asset owners are better off holding the factor—the market portfolio—than

¹ See Welch (2008) and Graham and Harvey (2001), respectively.

² Chapter 17 summarizes the history of index funds.